Literature and Evidence Review: Internal Enterprise Knowledge Assistants

Critical summary of the TVS Sidekick paper and related industry evidence, mapped to the Katalon Internal Assistant project

Unit	COS40005 Computing Technology Project A (September 2026 intake)
Project	Internal Assistant for Enterprise Knowledge (client: Katalon)
Client / Mentor	Katalon (mentor name to be confirmed)
Team	Group 3 (Project Manager: Truong Ngoc Gia Hieu)
Prepared	10 November 2026
Referencing	Harvard (Swinburne). All web sources accessed 10 November 2026; links are clickable.

1. Purpose and scope

This document records the evidence gathered to support the design of the Slack-integrated internal knowledge assistant being built for Katalon. It has two goals. First, it gives a critical summary of the primary academic source provided by the team, Reyero Lobo et al. (2025), which reports an in-house Retrieval Augmented Generation (RAG) assistant deployed at TVS Supply Chain Solutions UK. Second, it collects industry and vendor evidence relevant to the three project evaluation criteria: answer accuracy and relevance (40%), knowledge-base freshness (30%) and access-control correctness (30%). Because Katalon is now the client, it also records what Katalon has published about its own AI assistants, as client context.

2. Primary source: TVS Sidekick (Reyero Lobo et al., 2025)

Citation note. The uploaded file is named *2023_Dasgupta_TVS_Sidekick...*, but the paper's own metadata and title page list the authors as Reyero Lobo, Johnson, Buchanan, Shardlow, Williams and Attwood (TVS Supply Chain Solutions and Manchester Metropolitan University), and the arXiv identifier 2509.26482 dates it to 30 September 2025. It should therefore be cited as **Reyero Lobo et al. (2025)**, not Dasgupta (2023).

2.1 Context and motivation

TVS SCS UK is a third-party logistics provider in the UK and Europe. Rather than buying a vendor product, it chose to build its AI capability through an in-house team, arguing that external integrators would reduce agility and innovation. TVS Sidekick is that team's flagship product: an assistant that answers staff questions from internal company data, which employees reach through a Microsoft Teams extension (Reyero Lobo et al., 2025, p. 1). The paper claims three contributions: technical (architecture, prompt engineering and RAG), regulatory (aligning with the EU AI Act through ISO/IEC 42001) and sociotechnical (usage statistics and qualitative user feedback) (p. 2).

2.2 Technical architecture

The system has two components joined by a vector database (Reyero Lobo et al., 2025, pp. 4-5):

- **Ingestion pipeline.** Data is pulled from SharePoint, Azure DevOps, code repositories and the company website on a scheduled hourly refresh. Documents are parsed and chunked along semantic or logical boundaries, embedded, and stored in a separate index per data source. Source code is split into

functions or procedures, and an LLM writes a natural-language description of each file so that code can be retrieved and shown as a source.

- **RAG pipeline.** The query plus the conversation history from the last 60-minute session goes to a router. The router splits the query into sub-tasks and uses an LLM to choose a route; each route is a tool-based 'chatterbot' specialised for one data source. Each task runs retrieval (top-10 similar chunks), augmentation (parameter extraction and prompt template) and generation. The partial answers are then merged into one reply, and the query and response are logged.

2.3 Governance and regulation

The company adopted an AI Management System based on ISO/IEC 42001 and mapped its requirements onto management systems it already ran, notably ISO/IEC 27001 for information security. That mapping identified four focus areas: AI risks, monitoring and measuring, continuous improvement, and awareness and training (Reyero Lobo et al., 2025, pp. 5-6). The monitoring system tracks interaction volume, response time and user engagement, broken down by department, job title, user and question type (p. 6).

2.4 Evaluation findings

Evidence type	Key findings (Reyero Lobo et al., 2025, pp. 6-8)
Usage monitoring (Mar-Jun 2025)	Over 500 prompts in the first two months and about 250 in the next two. Conversations were short, rarely above an average of five questions. Response time peaked on certain dates, raising the average to 46 s. The most active users were in technology and business roles. Questions answered from SharePoint were the most common query type.
User segments	Clear split between lead or early adopters (46-253 queries each) and occasional users (fewer than 20 on average).
Qualitative interviews (n = 24, IT dept., Mar-Apr 2025)	Thematic analysis by two independent coders found: enhanced retrieval specific to company data (16 mentions); good at extracting business logic (10); not enough technical detail, especially for legacy code (13); keen to engage with AI (11); privacy, commercial-sensitivity and job-security concerns (8). Participants accepted the tool provided it used the right information and access levels.

2.5 Critical appraisal

- **Strengths.** It is a rare peer-authored account of a *production* internal assistant. It combines quantitative telemetry with structured qualitative research, and it shows how to turn regulation into concrete management-system controls.
- **Limitations.** The authors state that the findings come from one organisation over a short, early evaluation window (p. 8). No answer-accuracy metric (e.g., precision, groundedness, faithfulness) is reported, and permission enforcement is only raised as a user concern, not designed or tested. The interviewees all came from the IT department, which limits how far the results generalise. It is an arXiv preprint, so it should be described as not yet peer-reviewed.

3. Relevance to the Katalon project

The table compares TVS Sidekick with our brief for Katalon (Slack bot over Confluence using Amazon Bedrock) and lists implications for our design.

Dimension	TVS Sidekick (2025)	Katalon project	Implication for Group 3
Interface	Microsoft Teams extension	Slack bot	Same idea: meet employees in the chat tool they already use.

Dimension	TVS Sidekick (2025)	Katalon project	Implication for Group 3
Sources	SharePoint, Azure DevOps, code repos, website	Confluence spaces, documentation sites and forums (exact spaces to be confirmed with Katalon)	Use one index or metadata namespace per space, as TVS did per source.
Freshness (30%)	Scheduled hourly refresh	Background sync as Confluence changes	Hourly batch is a baseline; aim for incremental or event-driven sync and measure lag.
Accuracy (40%)	Router, per-source tools, top-10 retrieval; no accuracy metric reported	Graded on accuracy and relevance	Close this gap: build a labelled test set and report groundedness and citation correctness.
Access control (30%)	Raised as a user concern only	Answer only from documents the user may read	Our main contribution beyond TVS; see Section 5.
Governance	ISO/IEC 42001 AIMS, usage monitoring, interviews	Software company holding customer and product data	Adopt the monitoring metrics (volume, latency, engagement) and a feedback channel.

4. Industry case studies of internal knowledge bots

These sources are engineering blogs and case studies, not peer-reviewed research. They should be presented as industry evidence that the architecture is practical, not as proof of effectiveness.

Organisation	What they built	Relevance	Source
Aircall	Internal knowledge agent answering from Confluence, reached through a Slack bot; custom Confluence scraper storing data in S3; Amazon Bedrock guardrails; Confluence page URLs returned in answers.	Closest match to our stack (Slack + Confluence + Bedrock).	Aircall (n.d.)
Delivery Hero	'EmployeeGPT', a RAG bot built into Slack that now serves many teams and regions.	Shows a Slack RAG bot can scale across a large organisation.	Delivery Hero (n.d.)
KINTO Technologies	Internal Slack chatbot on AWS Lambda with RAG search over Confluence pages holding internal rules and procedures.	Serverless AWS pattern for a Slack bot.	Torii (n.d.)
System1	Slack bot over Confluence docs for engineering, management and science teams, using RAG on Snowflake Cortex.	Multi-team Confluence RAG bot.	Baker and Reardon (n.d.)
Nubank	'AskNu', RAG-based employee knowledge system for a fintech with about 9,000 employees in several countries.	Shows the pattern at large-company scale.	ZenML (n.d.)
Blue Reply (thesis)	Design, development and evaluation of an internal-knowledge assistant at an Italian IT consultancy.	Academic evaluation method to borrow.	Manieri (n.d.)

5. Evidence for access-control design (30% criterion)

- **Threat is recognised by standards bodies.** According to Glean (n.d.), the OWASP Top 10 for LLM Applications (2025) ranks sensitive information disclosure as the second-highest LLM risk and calls for permission-aware vector databases. (This is a secondary citation; the team should check the OWASP

document directly before the final report.)

- **Why naive RAG leaks.** Wavect (n.d.) explains that embedding drops a document's access-control list, so pure top-k similarity search retrieves by similarity, not permission. Its fix is to tag each chunk with ACL metadata at index time and filter on it at query time.
- **Bedrock-specific option.** Amazon Web Services (n.d.) contrasts the common replicate-and-filter approach with real-time ACL checks against the source system at query time, using Amazon Quick and Bedrock Knowledge Bases. This is the most relevant reference for our stack.

Design implication. Store Confluence space and page restrictions as chunk metadata, map the Slack user ID to a Confluence identity, apply a deny-by-default filter *before* chunks reach the LLM, and test with users who have different permissions.

6. Evidence for knowledge-base freshness (30% criterion)

- Amazon Web Services (2026) announced on 9 September 2026 that Amazon Bedrock Managed Knowledge Base now supports Confluence Data Center as a native data-source connector. The team should confirm with the Katalon mentor whether Katalon runs Confluence Cloud or Data Center, because that decides which connector applies.
- TVS Sidekick refreshes hourly (Reyero Lobo et al., 2025, p. 4), and Aircall (n.d.) uses a custom scraper into S3. Both are baselines against which our sync latency can be compared.

7. Client context: Katalon's existing AI assistants

No public source found shows that Katalon runs an internal, employee-facing knowledge bot. Its published assistants serve *customers*. The Katalon Support Assistant is a 24/7 AI support chat covering the whole product suite (Katalon, n.d.). The Katalon AI Assistant (formerly StudioAssist) is built into Katalon Studio and has an Agent mode that connects to Katalon documentation and project data through MCP servers (Katalon, 2026a; 2026b). These can be cited as examples of documentation-grounded assistants. This matters for the project in two ways. First, it positions our bot as a new, employee-facing tool that complements Katalon's customer-facing assistants, not a duplicate of them. Second, Katalon already uses MCP servers to ground its assistant in documentation, so an MCP-based design for our bot would fit the client's existing practice; this should be confirmed with the mentor. Any claim about tools Katalon uses internally should come from the mentor and be cited as a personal communication.

8. Summary

Reyero Lobo et al. (2025) confirm that an in-house RAG assistant inside the corporate chat tool is workable and valued by staff, and they give concrete patterns for ingestion, routing and monitoring. The paper leaves open the two areas that make up 60% of our assessment: measured answer accuracy and enforced document-level permissions. The industry and AWS sources above give a basis for addressing both, which positions our Katalon project to go beyond the published case.

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Note on dates: 'n.d.' marks pages that showed no publication date in the retrieved version. Before submission, open each link and add the date if one is shown. Lewis et al. (2020) is included because the TVS paper builds directly on it.